

Same Budget, More Water

Ranking Egypt's on-farm modernisation budget by the water that evaporated without growing a crop

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Abstract

FAO's WaPOR v3 database reports evaporation (E) and transpiration (T) as separate layers [1]. Transpiration is water that passed through a stoma and grew a crop. Evaporation is water that left the field having grown nothing. This work converts that distinction into an ordering over spending: for each administrative unit in Egypt's Nile Delta and Valley it estimates the volume of non-beneficial evaporation that is recoverable by low-cost on-farm measures, the annualised cost of recovering it, and hence a cost per cubic metre. Spending a fixed budget down that ordering recovers 25 % more water than distributing the same budget in proportion to irrigated area. The work is designed to sit immediately downstream of IPAT, the irrigation assessment tool already operating on the Egyptian Ministry of Water Resources and Irrigation platform [2], which reports *where* performance is poor but not *where to spend*. Two findings contradicted the authors' prior hypothesis and are reported in full, together with a self-audit of how much of the result is attributable to the satellite rather than to coarse agronomic knowledge. One substantial methodological objection is set out and addressed rather than omitted: WaPOR's E/T separation may be insufficiently accurate for this use [3].

1 The decision, and the gap in the toolchain

Egypt allocates a fixed share of the Nile among roughly 110 million people, with agriculture consuming over 85 % of the national water allocation across approximately 4 % of the land area [4]. The Ministry of Water Resources and Irrigation (MWRI) is responsible for that allocation and, under the National Water Resources Plan 2017–2037, for rationalising water use and rehabilitating conveyance infrastructure [5]. It spends an annual on-farm modernisation budget and is accountable for what that budget delivers.

FAO's applications catalogue now lists more than twenty operational WaPOR tools [6]. Egypt has its own: IPAT is a web tool hosted on the MWRI platform, built on WaPOR v3 Level 3 (20 m) data covering roughly 1.5 Mha of the Middle and West Delta across sprinkler, drip and flood systems, co-developed with IHE Delft [2]. It reports where water productivity is high and low.

The gap is not data, and it is not accuracy. A planning officer with a fixed budget must decide *which* governorates receive it and in what proportion. No output in the existing chain is denominated in the units of that decision. The tool reports millimetres per dekad; the decision is made in currency, hectares and cubic metres.

2 Method

2.1 Seasonal aggregation

WaPOR v3 Level 2 dekadal rasters were summed over the 18 dekads spanning 1 May to 31 October 2024, the summer (*Nili/Sefi*) season. Two corrections are required and neither is optional. WaPOR stores dekadal

values as scaled integers with a scale property of 0.1, and the scaled quantity is a rate in mm d^{-1} , not a total per dekad. Dekads are 10, 10, and 8–11 days long, so

$$X_{\text{season}} = \sum_{d=1}^{18} \text{DN}_d \times s \times n_d, \quad s = 0.1, \quad n_d = \text{days in dekad } d. \quad (1)$$

Because $sn_d \approx 1$, the raw integer coincidentally reads as millimetres per dekad, and a seasonal total computed with *neither* correction is also superficially plausible. The raw value is therefore not a usable diagnostic; only the seasonal total distinguishes the two cases.

2.2 Separating cropland from desert, and from water

Outside the irrigated Nile system there is effectively no water and no rain, so seasonal evapotranspiration collapses towards zero. A seasonal threshold of AETI > 200 mm therefore separates irrigated land from desert using the same data already being analysed, with no additional product, licence or cost.

Open water is a more serious problem, because it evaporates freely all season and would masquerade as extremely inefficient cropland. Three filters are applied in combination: JRC Global Surface Water occurrence above 50 % [7]; a beneficial-fraction floor of $T/\text{AETI} > 0.15$, below which a pixel cannot be a crop; and an absolute ceiling of $E < 600$ mm per season. Section 6 explains why the third filter proved necessary.

2.3 From evaporation to an ordering over spending

For each administrative unit i and applicable intervention j :

$$V_{ij} = E_i \cdot A_i \cdot 10 \cdot r_j \cdot m_j \quad [\text{m}^3 \text{yr}^{-1}] \quad (2)$$

$$C_{ij} = \frac{c_j}{L_j} \cdot A_i \quad [\text{USD yr}^{-1}] \quad (3)$$

$$\kappa_{ij} = C_{ij}/V_{ij} \quad [\text{USD m}^{-3}] \quad (4)$$

where E_i is mean seasonal evaporation over masked cropland (mm), A_i the observed irrigated area (ha), r_j the evaporation-reduction fraction, m_j the number of seasons per year over which the measure acts, c_j the capital cost per hectare and L_j the service life. Pairs are sorted by κ ascending and funded until the budget is exhausted. Because a unit may be partially treated, the greedy solution is optimal.

2.4 Comparator

The baseline distributes the same budget in proportion to irrigated area, using the same one-measure-per-unit rule so that the constraint cancels. This is a *neutral prior*, meaning what one would do knowing only where the cropland lies. It is explicitly not a claim about current MWRI practice, for which no citation is available. Against a literal equal split the reported advantage would be considerably larger; quoting that number would be comparing against a weaker counterfactual than honesty permits.

2.5 Intervention coefficients

The coefficient for alternate wetting and drying warrants comment. The saving usually attributed to AWD in the literature is in water *applied*, that is percolation and seepage, rather than in evapotranspiration [8]. In a closed, high-reuse system such as the Nile Delta, percolation is largely recovered downstream and therefore is not a saving at basin scale [9, 10]. Only the evaporation component is counted here, which is why $r_j = 0.10$ rather than the 0.25–0.30 commonly quoted. Two of the four measures are never selected at any budget between \$6M and \$80M; they are retained for completeness, not because they do work.

Table 1: Intervention model. These are literature-order estimates, not measurements, and every value is an editable input in the delivered tool.

Measure	USD ha ⁻¹	Life, yr	r_j	m_j	Note
Laser land levelling	75	5	0.10	2	Deployed at scale in Egypt
Alternate wetting & drying	25	1	0.10	1	Evaporation component only
Straw / plastic mulching	260	1	0.30	1	Never selected
Drip retrofit	1800	10	0.45	2	Never selected

3 Results

Observed irrigated area totalled 3.56 Mha against 3.21 Mha of reported cultivated area, an 11 % overshoot, in the expected direction for a satellite that also observes orchards, canal banks and reclamation the statistics omit. As an internal check, $E + T + I$ summed to AETI exactly, confirming the scale and day-count corrections.

Table 2: Allocation of a \$12M budget. Water figures are annual.

	Governorate	Funded, \$M	Mm ³ yr ⁻¹	USD m ⁻³
1	Aswan	1.19	29.5	0.040
2	Qena	2.74	67.8	0.040
3	Sohag	2.17	50.8	0.043
4	Assiut	2.23	48.3	0.046
5	Beni Suef	1.97	41.7	0.047
6	Qalyubia	1.12	22.9	0.049
7	Minya	0.58	11.8	0.049
	Targeted	12.00	273.0	0.044
	Area-proportional baseline	12.00	218.1	0.055

Ordering the same budget by cost of recovered water yields 25.2 % more water than distributing it by irrigated area, and the same figure on unit cost, so the gain is not an artefact of spending more.

3.1 Equity is priced rather than assumed away

Strictly cheapest-first funds seven governorates out of eighteen. No ministry operating under an equity mandate would sign that, so a recommendation which ignores the constraint is a recommendation which is ignored. Instead, a fraction of the budget is reserved and distributed across every unit by irrigated area before the remainder is targeted, and the cost of that choice is quantified (Table 3).

Table 3: The cost of an equity floor. A floor of 1.0 reproduces the baseline exactly, which serves as an internal consistency check.

Floor, %	Units funded	Mm ³	Cost vs. pure targeting
0	7	273.0	n/a
25	18	264.2	−3.2%
50	18	253.3	−7.2%
75	18	237.9	−12.9%
100	18	218.1	−20.1%

Reserving a quarter of the budget funds every governorate at a cost of 3.2 % of the water. This is offered as the practical recommendation: it does not depoliticise the allocation, it prices the political constraint and returns the dial to the committee.

4 Two findings that contradicted our hypothesis

4.1 The rice belt is the most efficient land in Egypt, not the least

The work was built expecting the northern Delta rice belt to lead, reasoning that ponded paddy evaporates throughout the season. The measurement says the opposite. Damietta returns an evaporated share of 0.157 and Kafr El Sheikh 0.191, against 0.23–0.29 elsewhere. A closed rice canopy shades the water surface, so ponding does not imply evaporation.

4.2 The ranking follows throughput, not practice

Table 4: Area-weighted evaporated share by irrigation system.

System	Irrigated, Mha	AETI, mm	$E/AETI$
Delta, rice belt	1.76	592	0.210
Nile Valley (Upper Egypt)	1.19	684	0.241
Delta, other	0.61	500	0.243

The wasted *fraction* is close to uniform nationally. What differs is throughput: Upper Egypt passes 660–775 mm through a field against the Delta’s 510–640 mm, so an equivalent inefficiency costs more water. Targeting therefore follows evaporative demand rather than poor practice. That is a materially different policy message from the one we set out to deliver.

5 Self-audit: how much of this requires the satellite?

The allocation was re-run with the satellite information progressively removed.

Information available	Gain vs. baseline, %
Real per-district evaporation (WaPOR)	25.2
Per-system average only (agronomic map)	21.5
Spatially uniform evaporation	0.0

With no spatial information, targeting gains nothing: the entire advantage is spatial. But most of it resides in the system-level pattern, so the increment attributable to district-level satellite *detail* is approximately 3.7 percentage points. This is a weaker claim than anticipated and is reported because a reviewer would otherwise derive it. The counter-argument is that the system-level pattern is precisely what the satellite corrected: without it the budget would have been directed into the Delta on the strength of a textbook intuition about ponded rice.

6 Contamination we introduced and then removed

Aswan initially ranked first nationally. Inspection of the underlying pixel distribution showed seasonal E reaching 1834 mm, with 5 392 pixels above 600 mm, which are rates attainable by open water and not by cropland. Lake Nasser, the Toshka lakes, the Nile channel, main canals and Delta aquaculture had survived both the JRC mask, whose occurrence baseline misses fluctuating margins and recent water bodies, and the beneficial-fraction floor, which a partially vegetated shoreline pixel can clear.

The effect was not marginal: Aswan’s mean E was 274 mm against a median of 192 mm. The mean was being set by contamination, and it alone had placed Aswan at the head of the national ranking. A physical ceiling of $E < 600$ mm was introduced. For every unit the 99th percentile of cropland E lies below 400 mm while open water exceeds 1200 mm, so nothing real occupies the interval. This reduced Aswan’s mean to 186 mm and largely removed its lead. The national advantage fell from 32.2% to

25.2 %, and the system-level contrast that had appeared to separate Upper Egypt from the Delta largely dissolved.

This correction is reported because the uncorrected result would have been more favourable to the hypothesis being tested. The uncorrected figures were the ones that best supported our own argument, which is the reason they needed checking first.

7 Limitations

7.1 The principal methodological objection

Chukalla et al. [3], in the reference framework for WaPOR-based irrigation performance assessment, *excluded the beneficial fraction* on the grounds that WaPOR's separation of evaporation from transpiration is insufficiently accurate. That is a direct challenge to the present work, which rests entirely on that separation, and it is not dismissed here.

Three considerations bear on it. First, the *E/T* split is used *relatively*, to order units, rather than absolutely; a systematic bias in the retrieval partly cancels under comparison between units processed by the same algorithm. Second, the self-audit above shows the ordering is driven substantially by AETI, the best-constrained WaPOR variable, through the throughput effect rather than by the fraction alone. Third, and decisively, the appropriate status of this output is a *prioritisation hypothesis to be confirmed by field measurement*, not a settled allocation. The correct response to Chukalla et al. [3] is to direct field validation at the units this method ranks highest, which is itself a use of the ranking.

7.2 Other limitations

- **Field scale is not basin scale.** A share of the water an intervention frees was already returning to drains and aquifers and being reused downstream [9, 10]. Suppressing soil evaporation may also convert it to transpiration and yield rather than to recoverable water. Recoverable evaporation is an upper bound; these figures rank locations and do not constitute a water account.
- **Magnitude is soft, ordering is comparatively robust.** The ordering follows observed evaporation; the size of the prize moves with every coefficient in Table 1.
- **The decision unit is administrative.** Governorates are not the delivery unit; irrigation directorates and command areas are. The method is boundary-agnostic and runs unchanged on any polygon set, including those used by IPAT.
- **Units below 5 000 ha are excluded** and reported, not silently dropped. GAUL's Luxor polygon covers 12 km², which is the city rather than the governorate [11].
- **The confidence indicator is presently unsound.** It is computed over the already-masked subset, so a predominantly desert unit can attain a narrow distribution and therefore a high score. It is reported in the tool and should not be relied upon.

8 Implementation

The analysis runs either as a local Python pipeline over exported GeoTIFFs or entirely within Google Earth Engine [12], in which case per-unit statistics are precomputed to an Earth Engine asset and the allocation is evaluated client-side. The delivered artefact is a single self-contained HTML file requiring no network, which matters because the intended user opens it inside a ministry rather than on a research network. All intervention coefficients are editable at runtime, so the committee argues with the model rather than being handed its conclusion.

Monitoring closes the loop. Because the indicator that justifies the spending is the indicator that measures it, the following season observes funded and unfunded units under the same instrument at the

same cost, permitting a difference-in-differences estimate with a stated confidence interval, including the case in which the effect is too small to detect.

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